

# Arthur K. MacKeith

## Curriculum Vitae

Contact information and digital portfolio at [Arthur.Mackeith.info](http://Arthur.Mackeith.info)

### EDUCATION

*Ph.D., Mechanical Engineering & Materials Science, Yale University* Expected May 2027

Advisor: Professor Corey S. O'Hern

Research topics:

- Modeling morphogenesis of leaf spongy mesophyll cells in three dimensions using molecular dynamics (MD) simulations with high resolution cell shape tracking
- Quantifying cell shape change using image analysis in 3D leaf mesophyll and 2D epithelial tissue
- Efficient methods for simulating deformable particles in MD simulation using CUDA and MATLAB

Honors and Awards:

NIH Biophysics Trainee 2022-2024

Program in Physics, Engineering, and Biology Travel Award, 2026, 2025, 2024, 2023

Yale Conference Travel Fellowship 2025

*M.S. & M.Phil. en route to Ph.D.*

May 2024

Grades: 8 Credits Honors, 2 Credits High Pass (Yale does not report a GPA)

Selected Course Work: Mathematical Methods of Physics, Statistical Physics I & II, Biological Physics, Biomedical Image Processing & Analysis.

*M.S., Computer Science, The University of Chicago*

September 2016 to June 2020

*B.A., Physics, The University of Chicago*

Joint degree with M.S.

Joint degree GPA: 3.7/4.0,

Selected Course Work: Computer Science: Math Foundations of Machine Learning, Introduction to Machine Learning, Algorithms, Computer Architecture, Parallel Programming, Time Series Analysis and Stochastic Processes.

Physics: Honors Mechanics, Intermediate Mechanics, Mathematical Methods of Physics, Geophysical Fluid Dynamics I and II.

Honors and Awards:

James Franck Institute Summer Research Fellowship (2018)

Dean's List (2016-2019)

### PUBLICATIONS

5. **Arthur K. MacKeith**, Dong Wang, Mark D. Shattuck, and Corey S. O'Hern. "Deformable Particles: Modeling and Applications." In *Packing Problems in Soft Matter Physics*, edited by Ho-Kei Chan, Stefan Hutzler, Adil Mughal, Corey S. O'Hern, Yujie Wang, and Denis Weaire, vol. 27. Theoretical and Computational Chemistry Series. Royal Society of Chemistry, 2025.

4. Sven Witthaus, Atoosa Parsa, Dong Wang, Nidhi Pashine, Jerry Zhang, **Arthur K. MacKeith**, Mark D. Shattuck, Josh Bongard, Corey S. O'Hern, and Rebecca Kramer-Bottiglio. "Evolution of adaptive force chains in reconfigurable granular metamaterials." *Soft Matter* 21, no. 30 (2025): 6088-6099.
3. Andrew T. Ton, **Arthur K. MacKeith**, Mark D. Shattuck, and Corey S. O'Hern. "Mechanical plasticity of cell membranes enhances epithelial wound closure." *Physical Review Research* 6, no. 1 (2024): L012036.
2. Won-Kyu Lee, Daniel J. Preston, Markus P. Nemitz, Amit Nagarkar, **Arthur K. MacKeith**, Benjamin Gorissen, Nikolas Vasios, Vanessa Sanchez, Katia Bertoldi, L. Mahadevan, and George M. Whitesides. "A buckling-sheet ring oscillator for electronics-free, multimodal locomotion." *Science Robotics* 7, no. 63 (2022): eabg5812.
1. Kieran A. Murphy, **Arthur K. MacKeith**, Leah K. Roth, and Heinrich M. Jaeger. "The intertwined roles of particle shape and surface roughness in controlling the shear strength of a granular material." *Granular Matter* 21, no. 3 (2019): 72.

## PRESENTATIONS

4. **Contributed:** Arthur K. MacKeith, Sam Banks, Adam B. Roddy, Mark D. Shattuck, and Corey S. O'Hern. "Mesophyll Development in 3D: Pressure, Plasticity, and Porosity." *American Physical Society: Global Physics Summit*. 2026.
3. **Invited:** Arthur K. MacKeith, Adam B. Roddy, Mark D. Shattuck, and Corey S. O'Hern. "Modeling Mesophyll Tissue Morphogenesis in Three Dimensions." *Gordon Research Seminar: Interdisciplinary Approaches to Soft and Living Matter*. 2025.
2. **Contributed:** Arthur K. MacKeith, Adam B. Roddy, Mark D. Shattuck, Corey S. O'Hern. "Modeling Spongy Mesophyll Development." *American Physical Society: Global Physics Summit*. 2025.
1. **Contributed:** Arthur K. MacKeith, Allison E. Culbert, John D. Treado, Adam B. Roddy, Craig Brodersen, Mark D. Shattuck, and Corey S. O'Hern. "Modeling and Analysis of Mesophyll Tissue Development in Leaves and Flowers Across Species." *American Physical Society: March Meeting*. 2023.

## CONTRIBUTIONS TO FUNDED PROPOSALS

2. NSF CMMI 2414268 "Development of an in Silico Full Leaf Model Validated by Experiments" Duration: September 1, 2024 to August 31, 2027
1. Yale Internal AI Course Revision Grant "Integrated Workshop for the Integrated Graduate Program in Physics, Engineering, and Biology" Duration: August 1, 2025 to June 30, 2027

## UNDERGRADUATES MENTORED

Evan Cook (Summer 2023), subsequently research fellow at Stanford RegLab

George Ploumis (2023-2024), subsequently Physics PhD student at Stanford University

Alexander Morand (2023-2024), subsequently Physics PhD student at Yale University

Alec Chen (2024-2025), subsequently Mechanical Engineering PhD student at Princeton University

Lucas Dong (Summer 2025)

David Conway (Summer 2026)

Marvid Odobashi (2024-2027)

## RESEARCH EXPERIENCE PRIOR TO PHD

*Robotic Intelligence through Perception Lab, Toyota Technological Institute at Chicago, Chicago IL*

Full Time Research Engineer

August 2019 to June 2021

- Optimized a finite element modeling (FEM) environment for reinforcement learning for soft robotics
- Integrated a low-cost drone into education, and research robotics platform Duckietown. These drones are currently being used in public high schools across Rhode Island for hands on STEM education

*George M. Whitesides Group, Harvard University, Cambridge MA*

Summer Research Experience for Undergraduates Student

June 2019 to August 2019

- Designed and built a low-cost semi-soft beam climbing robot, incorporating soft controls such that it only requires a single constant input pressure gas source to climb
- Designed a pneumatic digital to analogue signal processing demonstration and characterized it using MATLAB

*Jaeger Lab, The University of Chicago, Chicago IL*

Undergraduate Research Assistant

December 2017 to June 2019

- Wrote and optimized image analysis python program that fit the position and orientation of the several thousand particles in a granular sample with 99.5% success rate
- Analyzed position and orientation data in python for global trends and mesoscale structures
- Designed 3D particles to print that break predictably to detect shifts in packing structure
- Recorded and reconstructed computed tomography scans of packings of granular media

## OUTREACH

*Girls' Science Investigations, Yale Pathways, 2022-Present - Volunteer*

*Duckietown Robotics Teaching Platform (See Research Experience) - Developer 2019-2021*

*Physics with a Bang! Holiday Lecture and Open House, 2018 - Volunteer*

## SKILLS

- Proficient Languages: Python, C++, MATLAB, CUDA
- Languages with some experience: C, Go, DrRacket (LISP), MIPS Assembly Language
- Tools and Operating Systems: ROS, Docker, Sofa Framework (Finite element method software), Solidworks (some experience), Swig (C and C++ wrapper for Python and other languages), Ubuntu, OSX
- Hardware: 3D printing, Silicon molding, Soldering, Woodworking, Instron